

jailing the negligent driver, some lawyers would contend that fault lay with the car manufacturer. Realising this, carmakers are not offering the technology for sale.

The same legal issue prevents automatic braking in vehicles to prevent collisions. Cars like the Mercedes-Benz S-Class have radar that scans for possible collisions, and use electrically actuated brakes that could stop short of most collisions. But if the vehicle does strike an abruptly appearing obstacle, like another car that runs a red light at an intersection, the manufacturer can expect to be sued because the car didn't prevent the crash.

Cars combining telematics and GPS

The potential for future cars combining telematics and on-board navigation hardware is strong. Volkswagen, along with General Motors and other companies, is finding ways to let cars communicate their parking information directly to one another. A scenario where a car has crashed on a slippery road and comes to a rest in the lane, around a blind curve, is an example. In this instance, the crashed car could broadcast a local alert, warning any nearby cars of its presence to avoid a subsequent collision.

Car-to-car communications could be used to manage traffic signals and speed traffic flow, saving fuel and time.

The company has also collaborated with Google and nVidia to develop prototype navigation systems that collect map data from Google Earth, with 3D photorealistic images in place of the cartoonish maps of today's systems. Another benefit of getting map data from the Internet over a wireless link is that it is always up-to-date, in contrast to disk-based, on-board navigation systems, which are only as recent as the latest disk update.

Smart car charger

Automakers like General Motors, Toyota Motor and Nissan Motor are planning to bring rechargeable vehicles to



Engineer Michael Kintner-Meyer (front) and his team at Pacific Northwest National Laboratory developed the smart charger controller to manage peak demands in the electric grid once a mass of electric vehicles hit the road. The controller tells an electric vehicle's battery when to start and stop recharging based upon existing stress in the grid

the market as early as 2010. However, if a million owners plug their vehicles to recharge after work, it could cause a major strain on the power grid.

Researchers at the Pacific Northwest National Laboratory in Washington have come up with a way to recharge electric cars that won't strain the power grid. The smart charger controller automatically figures out the best and cheapest times for drivers to recharge cars. It communicates with the power grid and can temporarily stop charging when there is stress on the system.

Electric car owners could program the controller to charge at a specific time of day or night, such as midnight, rather than plugging right after work, when it's likely others would do the same, or when it may still be a peak period.

If time-based pricing is available, the device would wirelessly download the latest prices and decide when to charge based on the lowest costs.

The device can save consumers about \$100 a year by charging during off-peak hours.

Electric cars that recharge as they drive

Electric cars will now soon suck up energy as they travel over recharging

strips embedded in roadways. A South Korean university has developed a plan to power electric cars through recharging strips embedded in roadways that use a technology to transfer energy found in some electric toothbrushes. The plan, still in the experimental stage, will involve placing power strips about 20 cm to 90 cm wide and perhaps several hundred metres long built into the top of roads.

Cuttlefish-inspired TV screens

Two cuttlefish species can change their skin colour in response to the environment. Taking a cue, MIT scientists have developed an inexpensive but hi-tech screen that could be used in a variety of applications. Cuttlefish-inspired electronic ink and screens use less than one-hundredth the power of traditional television screens.

Cuttlefish change their colour by secreting different chemicals to change the spacing between membranes. MIT scientists created an artificial electrical system to control the spacing between layers, thereby changing the colours on the screen.

The current screen prototype is several square centimetres across but only one micron thick. Squeezed into that narrow space are 20 to 30 layers